

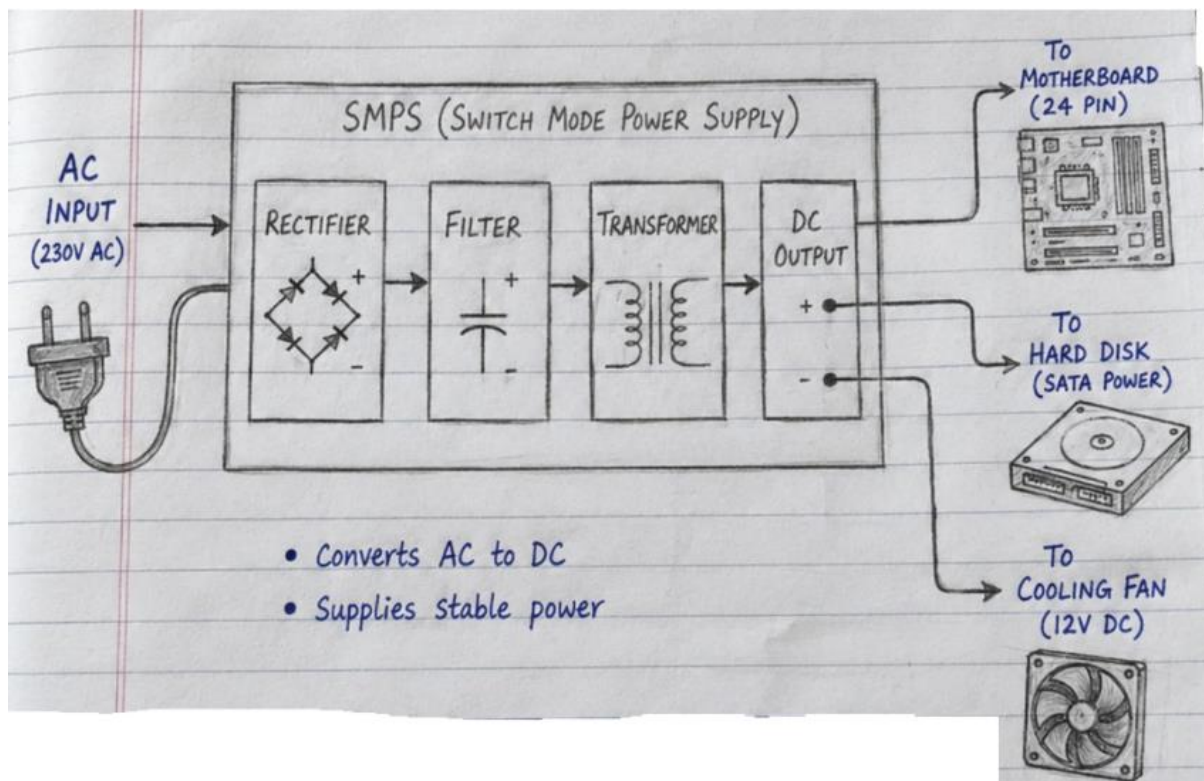
Assignment-2.2

“Powersupply system”

TASK-1 List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Supply Type	Device	WHY????
SMPS	Desktop Computer(PC)	Converts AC to multiple steady DC voltages (+3.3V, +5V, +12V) efficiently without excess heat.
SMPS	Laptop Charger brick	Provides high-efficiency power conversion in a compact, portable form factor.
SMPS	LED SmartTV	Delivers stable, regulated DC power to sensitive circuit boards and backlights.
UPS	Wi-fi router	Prevents internet drops during brief power cuts so online tasks aren't interrupted.
UPS	Desktop Computer	Provides emergency battery backup power to save work and shut down safely during outages.

TASK-2 Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. You can hand-draw and upload a photo, or use any free diagram tool like draw.



1. **AC Input (110V / 220V):** Draws raw alternating current from your wall outlet.
2. **Primary Rectifier & Filter:** Uses a diode bridge and large capacitors to convert high-voltage AC into high-voltage DC.
3. **Switching Element (Transistors):** Rapidly switches the DC voltage on and off thousands of times per second (kHz frequency) to regulate power efficiently.
4. **High-Frequency Transformer:** Isolates the high voltage and steps it down safely to low operating voltages.
5. **Output Rectifier & Filter:** Smooths the high-frequency pulsed power back into clean, stable DC voltage.
6. **DC Output Rails:** Distributes regulated power directly to PC components via dedicated voltage lines:
 - **+12V:** Powers high-draw components like the CPU and GPU.
 - **+5V:** Powers storage drives (SSDs/HDDs) and USB ports.
 - **+3.3V:** Powers motherboard logic chips and RAM.

TASK-3 Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.